

Rasika Rathnayake

**Teaching and Research Assistant,
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"An enthusiastic graduate student and researcher with a keen interest in researching molecular and cellular aspects of neurobiology with skills in communication, IT, and group working."

I am an inspirational, hardworking, organized, and creative individual with excellent communication and IT skills, working group ability, and willingness to achieve the best in research and academics with the utmost devotion to the entire scientific community. I am anxious to learn new professional skills in researching. Exhibits creative, well-organized, outstanding leadership experience, having served for numerous research projects and academic communities across the local and global scientific communities, with expertise in the planning, implementing, and publishing of the research and formulation of academic curricula. I am currently working as a teaching and research assistant with academic and research awards.

Research interests

As a self-motivated academic, I intend to achieve higher studies and research in Neurobiology, Cell biology, Bioinformatics, Nano-biotechnology, immunology, drug designing and synthesis, Biochemistry, System Biology, Enzymology, Phylogenetics, Genetics, microbiology, molecular medicine, and oncology.

Educational Qualifications:

- **Ph.D. Candidate in Biological Sciences:** Department of Biological Sciences, University at Buffalo, New York, The United States of America. **Running GPA 3.95/4.00**; May 2023 – Present
- **M.Sc in Biological Sciences:** Department of Biological Sciences, University at Buffalo, New York, The United States of America. **GPA 3.949/4.00**; May 2023

Key courses followed in Ph.D.: Cell Biology, Advanced Molecular Genetics, Human genetic diseases, Genes, and Cancers, Neurology and Integrative Neurology

Successfully completed two rotation projects under the supervision of Dr. Shermali Gunawardene and Dr. Paul Cullen as an initial requirement for the first-year Ph.D. Students and get enrolled in Dr. Gunewardene’s lab as a new Ph.D. student in March 2022. Qualified as a Ph.D. candidate on 10th of May and continuing Ph.D. under Dr. Shermali Gunawardene’s supervision.

- **BSc Four-year (Hons.) degree, Major: Molecular Biology and Biotechnology**, Faculty of Science, University of Peradeniya, Sri Lanka. **GPA 3.75/4.00 (First-class Honors)**; February 25, 2020

Key courses followed in BSc: Molecular Genetics, Biochemistry, Enzymology, Recombinant DNA Technology, Molecular Cell Biology, Bioinformatics, Molecular Immunology, Biotechnology, Molecular Biology of Plant and Animal Diseases, Molecular Evolution and Systematics, Biostatistics, Developmental Biology, Molecular Spectroscopic Techniques, Organic Chemistry, Inorganic Chemistry, Analytical Chemistry

Successfully conduct at 12 month-long undergraduate research project as partial fulfillment of the requirement on BSc on Pedimap: as a supportive tool in breeding decision making in Sri Lankan rice varietal improvement program and secure 4.00/4.00 of GPA under the supervision of Prof. S.D. Suneth S. Sooriyapathirana, University of Peradeniya, Sri Lanka.

- **Followed a Diploma in Journalism** at the Journalism Unit of the University of Colombo, Sri Lanka. (2014)
- **General Certification of Education (A/L)** (This is the qualification examination for university entrance); Physics A (pass > 75), Chemistry B (Very good pass > 65), Biology – C (Good pass > 55) (Department of Examinations, Sri Lanka, 2014)
- **General Certification of Education (O/L)** (This is the qualification examination for GCE AL); Buddhism A; Lang. & Lit (Sinhala) A; English A; Science & Technology A; Mathematics A; Soc. Stu. & History A; Art A; English Literature S; Health & Physical A. (Department of Examinations, Sri Lanka, 2007)

Professional Experiences

- **Teaching and Research Assistant** (August 19, 2021 – Present), Department of Biological Sciences, University at Buffalo, Buffalo, New York, The United States of America.

Nature of Duties

1. Working as a Research Assistant for Dr. Gunawardene’s Lab as a fourth-year Ph.D. candidate.
2. Assisting in experiments of undergraduate research projects and guiding them.
3. Supporting administrative activities in the lab.
4. Conducting laboratory classes in Biochemistry (BIO 315), Basic Physiology (BIO 313), and Evolution Biology (BIO 200) for 200 and 300-level undergraduates and assessing the undergraduate’s laboratory reports and assessments
5. Making and grading quizzes and assessments for undergraduate laboratory classes.

- **Teaching and Research Assistant** (March 12, 2020 – July 31, 2021), Department of Molecular Biology and Biotechnology, Faculty of Science, University of Peradeniya, Peradeniya, Sri Lanka.

Nature of Duties

1. Working as a Research Assistant in the Projects currently conducted by the department
2. Assisting in experiments of undergraduate research projects and guiding with thesis compiling
3. Supporting administrative activities
4. Teaching and guiding students from other faculties and universities who visit the department for Molecular Biology and Biotechnology-related lectures/practicals.
5. Conducting practical classes in Enzymology, Biochemistry, Recombinant DNA technology, Cell and Tissue culture, Bioinformatics, Molecular Phylogenetic, and Biostatistics for undergraduates and assessing the undergraduate’s laboratory reports and assessments

- **Teaching Assistant** (March 12, 2020 - July 31, 2021), Postgraduate Institute of Science, University of Peradeniya, Peradeniya, Sri Lanka.
MSc/MPhil/PhD Programs in Experimental Biotechnology
MSc/MPhil/PhD Programs in Nanoscience and Nanotechnology

Nature of Duties

1. Conducting practical classes for postgraduates’ and assess the postgraduate laboratory reports and assessments
2. Assisting and guiding experiments of postgraduate research projects
3. Supporting administrative activities

Communication Experiences	
• Worked as a Student announcer at Kandurata FM, Sri Kanka (2006 – 2010) • Worked as a presenter, producer, announcer and scriptwriter at Radio Kingswood – school radio station, Kingswood College, Kandy, Sri Lanka (2008 – 2010) • Worked as a trainer presenter at Stain Studios, Rathmalana, Sri Lanka (Sirasa TV) (2011)	
English Language Proficiency	
• TOEFL iBT test: Obtained an overall score of 87/120 for the TOEFL iBT test with individual band scores of Speaking: 25/30, Writing: 22/30, Reading: 22/30, and Listening: 18/30 (December 2020). • All the Ph.D. studies, TA, and RA duties are conducted in the English Medium • All the BSc studies were conducted in the English medium • Followed Extension Course in English, conducted by Faculty of Arts, University of Peradeniya, Sri Lanka. (2012) • Distinction ('A') passes for General English at the G.C.E. (O/L) Examination, 2007 and Very good pass ('B') for General English at the G.C.E. (A/L) Examination, 2014	
Appointments	
2022 – 2024	Graduate Student Ambassador for School of Art and Sciences , State University of New York at Buffalo.
2020	Member of the technical support team in the PGIS Research Congress technical Session, organized by the Postgraduate Institute of Science, University of Peradeniya, Sri Lanka
2020	Member of the teaching panel and the organizing committee for the workshop on 'Biology and Recombinant DNA Technology' for the AL (Biology) teachers, organized by the Board of Study in Science Education, Postgraduate Institute of Science, University of Peradeniya, Sri Lanka
2019 – 2020	Academic Assistant for designing of New MSc Program in Molecular Biology and Biotechnology , Postgraduate Institute of Science, University of Peradeniya, Peradeniya, Sri Lanka
Professional Services	
May 2026	Served as a reviewer for the Mark Diamond Research Fund (MDRF) Review Council , reviewing grant applications submitted for the Spring 2026 funding cycle, University at Buffalo.
Oct 2025	Served as a representative for the Department of Biological Sciences , University at Buffalo, at the WiSE/CSTEP STEM Research Opportunity Fair organized by the University at Buffalo.
Sep 2025	Hosted undergraduate students from Mercyhurst University , Erie, Pennsylvania, and introduced them to the research and academic environment of the Department of Biological Sciences, University at Buffalo.
2020- 2021	Academic assistant for designing and reviewing the B.Sc. Program in Applied Biology , Faculty of Applied Sciences, Southeastern University of Sri Lanka.

Computer literacy
- Completed the undergraduate research project which involved many data analyses procedures using computer software - Expert skills in graphic designing using Adobe Illustrator and CorelDRAW. - Expert skills in image editing and processing using Adobe Photoshop - Expert skills in 2D and 3D animation production using Abode Aftereffects, 4D Cinema, and 3D Max. - Trained skills in photography, videography, and video editing using Adobe Audition and Premiere. - Data representation and phenotyping analysis using Pedimap and FlexQTL. - Data management using MS office packages, the statistical packages SAS, Minitab, and R. - Bioinformatics for analyzing and retrieving molecular data. - Compilation of research project reports, writing research manuscripts, research proposal formulation, and standard formatting of seminar presentations, posters, and abstracts using MS office packages and Adobe InDesign.

Publications

Journal Articles
Rathnayake R., Vasanthan, J., Huang, J., Quiles-Gross, J., Lawvere, O., Bermudez, A., Bunk, S., Shabani V., and Gunawardena S. 2026. The phosphorylation state of huntingtin at S13/S16 and its effect on axonal transport and cell death. In-review - Krzystek T.J., Rathnayake R, Zeng J., Huang J., Iacobucci G., Yu, M.U., and Gunawardena, S. 2025. Opposing roles for GSK3β and ERK1-dependent phosphorylation of huntingtin during neuronal dysfunction and cell death in Huntington’s disease. <i>Cell Death & Disease</i>. (Accepted 2025 March) - Swinter, K., Salah, D., Rathnayake, R. and Gunawardena, S., 2023. PolyQ-Expansion Causes Mitochondria Fragmentation Independent of Huntingtin and Is Distinct from Traumatic Brain Injury (TBI)/Mechanical Stress-Mediated Fragmentation Which Results from Cell Death. <i>Cells</i>, 12(19), p.2406. - Krzystek, T.J., White, J.A., Rathnayake, R., Thurston, L., Hoffmar-Glennon, H., Li, Y. and Gunawardena, S., 2023. HTT (huntingtin) and RAB7 co-migrate retrogradely on a signaling LAMP1-containing late endosome during axonal injury. <i>Autophagy</i>, 19(4), pp.1199-1220. Rathnayake, PGRG., Sahibdeen, SM., Udawela, UAKS., Weebadde, CK., Weerakoon, WMW., Sooriyapathirana, SDSS. (2020) Application of Pedimap: a pedigree visualization tool to facilitate the decisioning of rice breeding in Sri Lanka. <i>Nature Scientific Reports</i> 10, 14255. https://doi.org/10.1038/s41598-020-71260-y Rathnayake, PGRG., Salih, R., Wijesundara, WWMUK., Sumanarathne, WGAS., Ranaweera, LT., Jayarathne, HSM., Weebadde CK., Sooriyapathirana, SDSS. (2020) Morphological variation, species delimits, and phylogenetic relationships of four important <i>Ipomoea</i> species in Sri Lanka. <i>Sri Lankan Journal of Agriculture and Ecosystems</i>, 2(1): 14-33. - Rathnayake, RMSK., Rathnayake, PGRG., Jayarathne, HSM., Nizam, R., Ranaweera, LT., Kannangara, SK., Udawela, UAKS., Weebadde, CK., Sooriyapathirana, SDSS. (2019) Assessment of the sequence-based haplotype-variants in selected DNA marker loci for the molecular breeding of resistant rice varieties to brown plant hopper. <i>Sri Lankan Journal of Agriculture and Ecosystems</i>, 1(1): 2-15. - Senevirathna, DGPK., Rathnayake, PGRG., Ranaweera, LT., Weebadde, CK., Sooriyapathirana, SDSS. (2020). Assessment of the species delimits and seasonal variation of fruit size, organoleptic preference

and phytochemical content of rose apple species in Sri Lanka. *Journal of Technology and Value Addition*. 2(2): 13-38.

9. Wijesundara, WMDA., Nakandala, NDUS., Wijesundara, WWMUK., Navanjana, PWI., Ranaweera, LT., Jayarathne, HSM., **Rathnayake, PGRG.**, Weebadde, CK., Wickkramaarachchi, WART., Sooriyapathirana, SDSS. (2020). Establishment of apple as an appealing cash crop in Sri Lanka: need, challenges, and opportunities. *Ceylon Journal of Science*, 49(4), 409–421. DOI: <http://doi.org/10.4038/cjs.v49i4.7821>
10. Jayarathne HSM., **Rathnayake, PGRG.**, Ranaweera, LT., Nizam, R., Rathnayake, RMSK., Kannangara, SK., Udawela, UAKS., Weebadde, WM., Sooriyapathirana, SDSS. (2020) Assessment on variation of performance indicators under phosphorus-starved conditions in a core-set of rice cultivars and their diversity in *pup1*-linked DNA marker-haplotypes. *Journal of the National Science Foundation of Sri Lanka*: Accepted.
11. Ranaweera, LT, Jayarathne, HSM., Thenuwara, G., **Rathnayake, PGRG.**, Karunarathne, SI., Thilakarathne, NK., Salih, R., Weebadde, CK., Weebadde, PC., Sooriyapathirana, SDSS. (2020) Photosynthetic phenomics of field and greenhouse grown amaranths vs. organoleptic preference and species delimits. *Plant Phenomics (AAAS partner journal)*: Accepted.
12. Ananda, GKS., Wijesundara, WWMUK., Herath, HMVPK., Karunarathne, SI., Kannangara, SK., Ranaweera, LT., **Rathnayake, PGRG.**, Abeysundara, SP., Udawela, UASK., Weerakoon, WMW., Weebadde, CK., Sooriyapathirana, SDSS. (2020) Establishment of a scheme of DNA fingerprinting and sequencing to distinguish the caryopsis samples of the exportable rice cultivars from a panel of mega-varieties. *Sri Lankan Journal of Agriculture and Ecosystems* 2(2): 31-55.

Review Articles

1. Rathnayake, RMSK., Karunarathne, SI., Ranaweera, LT., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS. (2020). Battling against brown planthopper in rice: genomic selection in backcross breeding to produce resistant cultivars - a review. *Sri Lankan Journal of Agriculture and Ecosystems* 2(2): 5-30.

Books

1. Jayasundara, EJMIK., Thivya B., Mahathantrige, HT., Ranatunga, RAS., **Rathnayake, PGRG.**, Warallanda, WGAP, Wijayasinghe, WMMM, Wijesekara, WWMLN, Salih, R., Senanayaka, SGMSD (2020) *Biosafety (A book in Sinhala Language)* (Editor: Sooriyapathirana SDSS). ISBN 978-955-41753-6-5. All authors have contributed equally, and their names are indicated according to the order of letters in Sinhala Alphabet.
2. Thivya, B., Salih, R., Senanayake, SGMSD, Mahathantrige, HT., Jayasundara, EJMIK., **Rathnayake, PGRG.**, WWWML, Warallanda., WGAP, Wijayasinghe., WMMM Nathali., Ranatunga, RAS. (2020). *Biosafety, Theory and application for GMO/GMF (A book in Tamil language)* (B. Thivya, R. Salih, J. Dhanushya, SDSS Sooriyapathirana Eds.). ISBN 978-955-41753-8-9

Book Chapters

1. Wijesundara, WWMUK., Salih, R., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS. (2020) Genetics, molecular biology and biotechnology as new revolutionary fields of science and technology. - Chapter 1. In Sooriyapathirana, S.D.S.S. (Ed.), *Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language)*. ISBN 978-955-41753-7-2. pp. 1-29.
2. Athukorala, NK., Karunathilaka, KMHL., Gunatilake, KS., Gunawardane, MHMIM., Chandrasekara, TM., Thilakarathne, BAAS., Piyawardana, MDSD., Perera, AYP., Makuldeniya, MWMTUG., Sunilrathne, VS., Suriyaarachchi, H., Warallanda, WGAP., Nathali, WWWML., **Rathnayake, PGRG.**, Salih, R., Thivya, B., Sooriyapathirana, SDSS. (2020). Principles and applications of genetics. – Chapter 2. In Sooriyapathirana, S.D.S.S. (Ed.), *Genetics, Molecular Genetics and Biotechnology (A textbook in*

Sinhala Language). ISBN 978-955-41753-7-2.) First-eleven authors have contributed equally – Each of them can be considered as the first co-author.)) These five authors have contributed equally – Each of them can be considered as the second co-author.

3. Athukorala, NK., Karunathilaka, KMHL., Gunatilake, KS., Gunawardane, MHMI., Chandrasekara, TM., Thilakarathne, BAAS., Piyawardana, MDSD., Perera, AYP., Makuldeniya, MWMTUG., Sunilrathne, VS., Suriyaarachchi, H., Mahathanthrige, HT., Senanayaka, SGMSD., **Rathnayake, PGRG.**, Salih, R., Thivya, B., Sooriyapathirana, SDSS. (2020) Molecular genetics: core-concepts and their importance. – Chapter 3. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Genetics and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.) First-eleven authors have contributed equally – Each of them can be considered as the first co-author.)) These five authors have contributed equally – Each of them can be considered as the second co-author.
4. Athukorala, NK., Karunathilaka, KMHL., Gunatilake, KS., Gunawardane, MHMIM., Chandrasekara, TM., Thilakarathne, BAAS., Piyawardana, MDSD., Perera, AYP., Makuldeniya, MWMTUG., Sunilrathne, VS., Suriyaarachchi, H., Ranatunga, RAS., Jayasundara, EJMIK., **Rathnayake, PGRG.**, Salih, R., Thivya, B., Sooriyapathirana, SDSS. (2020) Techniques of genetic engineering – recombinant DNA technology. – Chapter 4. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Genetics and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.) First-eleven authors have contributed equally – Each of them can be considered as the first co-author.)) These five authors have contributed equally – Each of them can be considered as the second co-author.
5. Thilakarathne, BAAS., Karunathilaka, KMHL., Athukorala, NK., Salih, R., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS (2020) DNA fingerprinting. – Chapter 7. In Sooriyapathirana, SDSS (Ed.), Genetics, Molecular Genetics and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
6. Perera, AYP., Suriyaarachchi, H., Makuldeniya, MWMTUG., Ranatunga, RAS., Nathali, WWWML., **Rathnayake, PGRG.**, Salih, R., Sooriyapathirana, SDSS. (2020) RNAi technology, gene therapy and gene editing using CRISPR-Cas9. – Chapter 9. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
7. Sunilrathne, VS., Piyawardana, MDSD., Suriyaarachchi, H., Senanayaka, SGMSD., Mahathanthrige, HT., **Rathnayake, PGRG.**, Salih, R., Sooriyapathirana, SDSS. (2020) Molecular biological methods associated with single nucleotide polymorphisms, microarrays and genotyping by sequencing. – Chapter 10. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Genetics and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
8. Gunatilake, KS., Makuldeniya, MWMTUG., Sunilrathne, VS., Nathali, WWWML., Ranatunga, RAS., Salih, R., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS. (2020) Nature of cancer and treatments. – Chapter 12. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
9. Karunathilaka, KMHL., Thilakarathne, BAAS., Piyawardana, MDSD., **Rathnayake, PGRG.**, Salih, R., Sooriyapathirana, SDSS. (2020) Breeding methods and plant micropropagation. – Chapter 15. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
10. Jayasundara, EJMIK., Thivya, B., Mahathanthrige, HT., Ranatunga, RAS., **Rathnayake, PGRG.**, Warallanda, WGAP., Nathali, WWWML., Salih, R., Senanayaka, SGMSD., Sooriyapathirana, SDSS. (2020) Applications of biotechnology and biosafety. – Chapter 16. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.) First-nine authors have contributed equally – Each of them can be considered as the first co-author.
11. Wijesundara, WWMUK., **Rathnayake, PGRG.**, Salih, R., Sooriyapathirana, SDSS. (2020) Introduction to bioinformatics. – Chapter 17. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.

12. Nakandala, NDUS., Salih, R., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS. (2020) Animal biotechnology. – Chapter 18. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
13. Ranatunga, APDT., Salih, R., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS. (2020) Marine biotechnology. – Chapter 19. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
14. Senevirathna, DGPK., Salih, R., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS. (2020) Environmental biotechnology. – Chapter 20. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
15. Sumanarathna, WGAS., Salih, R., **Rathnayake, PGRG.**, Sooriyapathirana, SDSS. (2020) Microbial biotechnology. – Chapter 21. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
16. Karunathilaka, KMA., **Rathnayake, PGRG.**, Salih, R., Sooriyapathirana, SDSS. (2020) Pharmaceutical biotechnology. – Chapter 22. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
17. Samarakoon, SMNS., **Rathnayake, PGRG.**, Salih, R., Sooriyapathirana, SDSS. (2020) Food and fermentation biotechnology. – Chapter 23. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.
18. Karunathilaka, KMA., Samarakoon, SMNS., Sumanarathna, WGAS., **Rathnayake, PGRG.**, Salih, R., Sooriyapathirana, SDSS. (2020) Industrial and Career Opportunities in Biotechnology as the Modern Revolution in Biology and Concluding Remarks. – Chapter 26. In Sooriyapathirana, SDSS. (Ed.), Genetics, Molecular Biology and Biotechnology (A textbook in Sinhala Language). ISBN 978-955-41753-7-2.) First-three authors have contributed equally – Each of them can be considered as the first co-author.

GenBank submissions

<https://www.ncbi.nlm.nih.gov/popset/1863294931> (*Ipomoees pes-caprae*: 3 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1722505244> (*Artocarpus* spp. 12 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1722505268> (*Artocarpus* spp. 12 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1722505280> (*Artocarpus* spp. 12 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1722505304> (*Artocarpus* spp. 12 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1722505316> (*Amaranthus* spp. and *Celosia cristata*: 6 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1722505328> (*Amaranthus* spp. and *Celosia cristata*: 6 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1789811523> (*Oryza sativa*: 27 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1789811496> (*Oryza sativa*: 27 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1864317285> (*Oryza sativa*: 15 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1864317334> (*Oryza sativa*: 5 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1864317416> (*Oryza sativa*: 5 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1864317378> (*Oryza sativa*: 5 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1819749450> (*Syzygium* spp.: 15 sequences)
<https://www.ncbi.nlm.nih.gov/popset/1819749465> (*Syzygium* spp.: 15 sequences)

Presentations and talks

1. **Rathnayake R.**, Vasanthan, J., Huang, J., Quiles-Gross, J., Lawvere, O., Bermudez, A., Bunk, S., Shabani V., and Gunawardena S. (2026) Identifying how S13/16 phosphorylation of huntingtin (HTT) influences axonal transport and cell death. **Biological Science Research symposium** organized by BGSA, Department of Biological Sciences, University at Buffalo; 15th January 2026.

2.

Rathnayake R., Vasanthan, J., Huang, J., Quiles-Gross, J., Lawvere, O., Bermudez, A., Bunk, S., Shabani V., and Gunawardena S. (2026) Identifying how S13/16 phosphorylation of huntingtin influences axonal transport and cell death. **18th Annual Neuroscience Research Day**, organized by Jacobs School of Medicine and Biomedical Sciences, University at Buffalo; 7^h November 2025

3.

Rathnayake R., Krzystek, T.J., Yu, M. and Gunawardena S. (2024) ERK phosphorylates the Huntington’s disease (HD) protein, huntingtin (HTT), and protects neurons against HD neuropathology. **17th Annual Neuroscience Research Day**, organized by Jacobs School of Medicine and Biomedical Sciences, University at Buffalo; 26th September 2024.

4.

Rathnayake R., Krzystek, T.J., Yu, M. and Gunawardena S. (2024) GSK3β and ERK kinases can phosphorylate the Huntington’s disease protein Huntingtin (HTT). **Biological Science Research symposium** organized by BGSA, Department of Biological Sciences, University at Buffalo; 16th May 2024.

5.

Identifying the role of ERK kinase in the phosphorylation of huntingtin protein in Huntington’s disease. GradCon - SUNY Graduate Research Conference 2024

6.

Rathnayake R. and Gunawardena S. (2023) Calcium-containing KDEL vesicles move bidirectionally within larval axons. Poster presentation at **Biological Science Research symposium** organized by BGSA, Department of Biological Sciences, University at Buffalo; 26th January 2023.

7.

Rathnayake R. and Gunawardena S. (2022) Identification of calcium-containing KDEL-vesicles within larval axons. Poster presentation at **Biological Science Research symposium** organized by BGSA, Department of Biological Sciences, University at Buffalo; 17th January 2022.

8.

Rathnayake R. and Gunawardena S. (2022) Calcium-containing KDEL-vesicles move bidirectionally within larval axons. Poster presentation at **SNFc Annual Meeting 2022** organized by Buffalo Society for Neuroscience Chapter, University at Buffalo; 28th October 2022.

Awards

•

Second Runner-Up, Best Talk, Biological Sciences Research Symposium, organized by the Biological Graduate Student Association, Department of Biological Sciences, University at Buffalo, January 15, 2026.

•

The Best Poster Award at the 3rd Graduate Research Conference, organized by the **State University of New York (2024), USA**.

•

The Peradeniya University Gold Medal Faculty of Science (2020) for **excellence in scholastic and extra-mural excellence**, University of Peradeniya, Sri Lanka

•

The Faculty Awards for Best Academic performance in Bachelor of Science, Faculty of Science, University of Peradeniya, Sri Lanka (2020).

•

University Award for the Academic Excellence (2020) for **the meritorious performance in B.Sc. program**, University of Peradeniya, Sri Lanka

•

D.F. Kariyakarawana Gold Medal for **Best Project Report** of the diploma in Journalism, University of Colombo, Sri Lanka (2014)

Grants and Fellowships

•

Mark Diamond Research Fund (MDRF), University at Buffalo — \$5,000 (May 2025–May 2026)

•

Departmental fellowship of 3900 USD per year from the Department of Biological Sciences, University at Buffalo, Buffalo, NY, The USA (2021-2023) for excellent academic background.

Professional training

•

Successful completion of a course on Bioinformatics which covered the sections on phylogenetic analysis and usage of NCBI, ExPASy, EMBL, and BLAST for analyzing and retrieval of molecular data (2018)

- Successful completion of the Scientific Writing course, which included the sections on project proposal writing, abstract writing, interpretation of results, and scientific manuscript writing
- Successful completion of a course on Biostatistics which covered the sections on hypothesis testing and data interpretation and usage of the statistical packages SAS, Minitab and R. (2019)
- Successful completion of a course on seminar presentation; conducted an oral presentation on "CTIP2- transcription factor as a therapeutic drug target against human medium spiny neurons degeneration in Huntington disease" and secured 4.00/4.00 GPA at Department of Molecular Biology and Biotechnology, Faculty of Science, University of Peradeniya, Sri Lanka

Technical skills related to Molecular Biology

- DNA extraction using CTAB method and kits, DNA band separation using Agarose and Polyacrylamide Gel Electrophoresis methods, Spectroscopy: UV/Visible
- Performing sanger sequencing for PCR products
- General Molecular Biology procedures including PCR methods
- Protein extraction and Purification methods, SDS-PAGE
- Microbiological methods for culturing and maintaining Bacterial and Fungal cultures
- Molecular systematics, phylogenetic analysis using MEGA, PoPArt, PAUP, and MrBayes software
- Performing statistical analyses using SAS and Minitab software

Extracurricular Activities

University of Peradeniya, Sri Lanka

- President of Molecular Biology and Biotechnology Association (2019 - Present)
- Secretary of Molecular Biology and Biotechnology Association (2018–2019)
- Committee member of Zoologists' Association (2017-2018)
- Member of the Music Society (2017-2018)
- Member of The Ceylon University Dramatic Society (2017-2018)

Kingswood College, Kandy, Sri Lanka

- President of Kingswood Media Network, Kingswood College, Kandy, Sri Lanka.
- President of Sinhala Literary Association of Kingswood College, Kandy, Sri Lanka.
- President of Kingswood Debate Circle (Sinhala), Kingswood College, Kandy, Sri Lanka.
- President of Buddhist Society of Kingswood College, Kingswood College, Kandy, Sri Lanka.
- President of PRATHIBA Art association of Kingswood College, Kingswood College, Kandy, Sri Lanka.
- Secretary of Astronomical Society of Kingswood College, Kingswood College, Kandy, Sri Lanka.
- Member of Kingswood Debate team, Kingswood College, Kandy, Sri Lanka.
- Member of Kingswood Photography Club, Kingswood College, Kandy, Sri Lanka.
- Member of Science Society, Kingswood College, Kandy, Sri Lanka.
- Member of English Literary Association, Kingswood College, Kandy, Sri Lanka.
- Member of Saukyadana Society (Sri Lankan non-governmental volunteer organization providing health education and first aid training) Kingswood College, Kandy, Sri Lanka.
- Member of English Drama Unit, Kingswood College, Kandy, Sri Lanka.
- Member of Electronic Society, Kingswood College, Kandy, Sri Lanka.
- Member of Psychological Association, Kingswood College, Kandy, Sri Lanka.

Referees	
Shermali Gunawardena <i>Assit. Professor</i> Neuro Science, Department of Biological Sciences, H540, State University of New York, Amherst, United States +858 212 2422 sg99@buffalo.edu	Prof. Paul J. Cullen <i>Professor</i> Biological Sciences, Department of Biological Sciences, H540, State University of New York, Amherst, United States +1 (716) 645-4923 picullen@buffalo.edu
Prof. R.G.S.C. Rajapakse <i>Professor</i> Molecular Biology and Biotechnology, Dean of the Faculty, Faculty of Science, University of Peradeniya, Sri Lanka. +94 81 239 (4505) + 94 77 780 1257 sanathrajapakse@yahoo.com sanathr@pdn.ac.lk	Prof. S.D.S.S. Sooriyapathirana <i>Senior Research Scientist</i> Molecular Biology and Biotechnology, PhotosynQ Inc. East Lansing, Michigan, United States +1 (517) 230-2343 sunethuop@gmail.com sunethss@sci.pdn.ac.lk

With this, I certify that the particulars submitted by me in this application are true and accurate. I am aware that if any of these particulars are false or inaccurate, I am liable to be disqualified before selection and dismissed without any compensation if the inaccuracy is detected after the appointment.

28th Ma7, 2026


Signature